

- 23** Bacteria were grown in a medium containing ^{15}N . After several generations, all of the DNA contained ^{15}N .

Some of these bacteria were transferred to a medium containing the common isotope of nitrogen, ^{14}N . The bacteria were allowed to divide once. The DNA of some of these bacteria was extracted and analysed. This DNA was all hybrid DNA containing equal amounts of ^{14}N and ^{15}N ($^{14}\text{N}/^{15}\text{N}$).

In another experiment, some bacteria from the medium with ^{15}N were transferred into a medium of ^{14}N . The bacteria were allowed to divide three times. The DNA of some of these bacteria was extracted and analysed.

What is the composition of this DNA?

